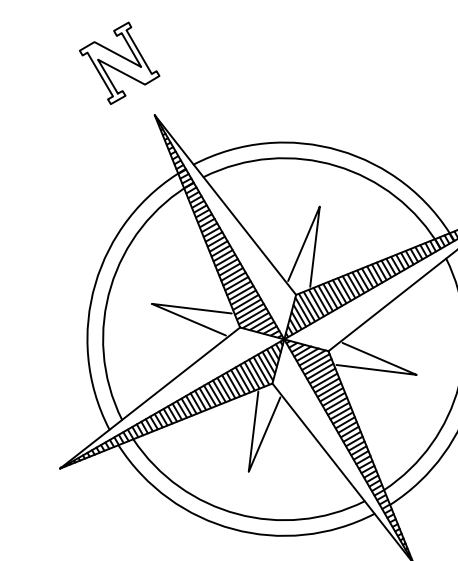
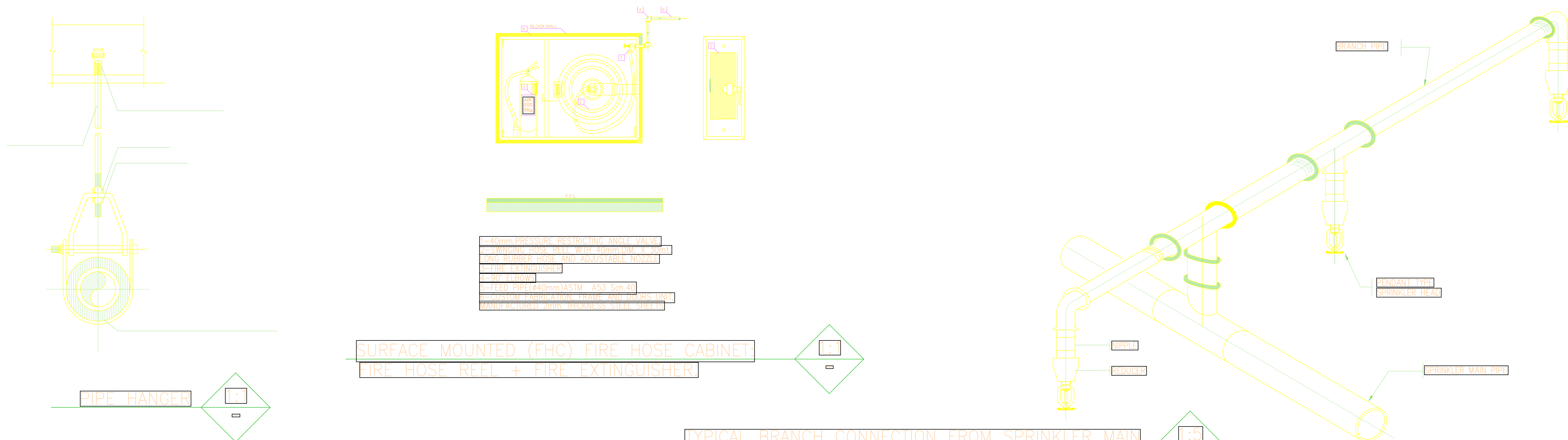
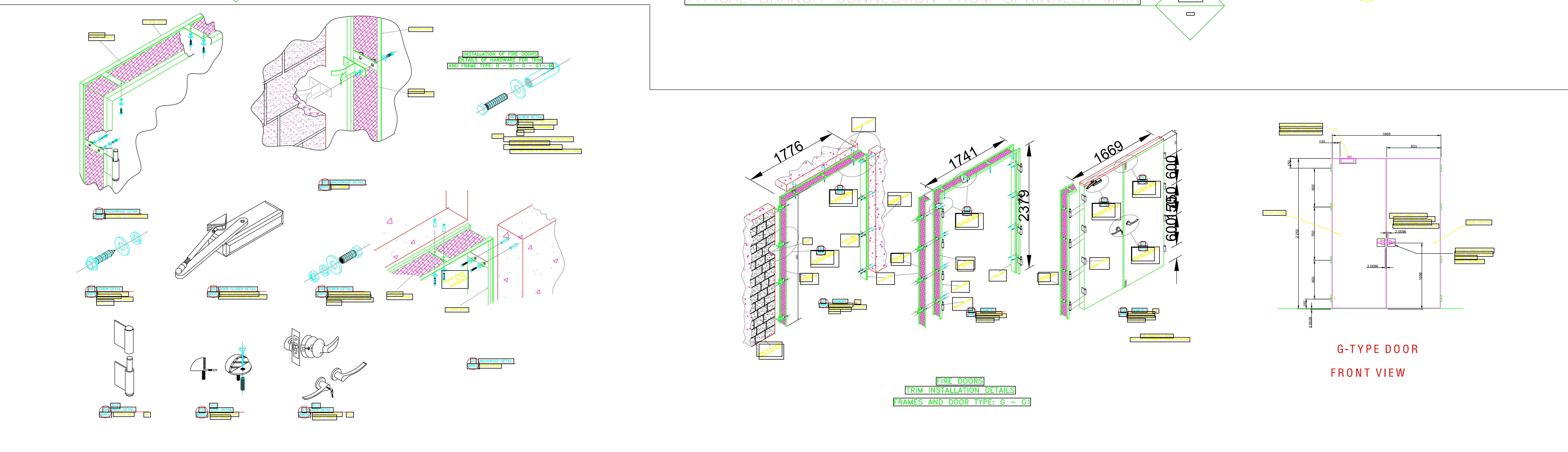




ALL UNITS IN METER



Design Studio (10), Technical
 Project, 2025/2026

INSTRUCTORS:

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DONE BY:
 ABDOLELAH ALQABBANI

PROJECT:
 Emergency Hospital

DRAWING NAME:
 DETAILS & CALCULATION

SCALE:
 1:150

SHEET NUMBER:
 A59

ALL COMPARTMENTS WATER DEMAND CALCLATION FOR FIRE RESVIER

Given Remote Area = 151 m²
 Density of flow = 4.0 L/min/m²
 Flow for Sprinklers = Area x Density
 = 151 m² x 4.0 L/min/m² = 604 L/min
 NO. Of sprinklers= 604 / 75 = 8.05 = 9 sprinklers

Required flow for Fire Fighting
 =Hose Stream Allowance +
 Flow for Sprinklers
 =379 L/min + 604 L/min = 983 L/min

Design Duration = 30 min

Volume of water required for fire fighting
 = Required flow x Design duration
 = 983 L/min x 30 min

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